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Experiment: 7

Questions:

Question 1: Reverse Words in a String.

Problem:

Write a function that takes a string and reverses each word individually, while keeping the word order the same.

Question 2: Find Duplicate Elements in an Array.

Problem:

Write a function to find and return all duplicate elements in a given array.

Question 3: Count Character Frequency.

Problem:

Write a function that counts the frequency of each character in a string and returns the result as an object.

Question 4: Merge Two Objects.

Problem:

Write a function to merge two objects. If a key exists in both objects, sum their values.

Question 5: Flatten Nested Array (One Level).

Problem:

Write a function to flatten a nested array by one level only.

Question 6: Find the Longest Word in a String.

Problem:

Write a function that returns the longest word from a given sentence.

Question 7: Group Anagrams.

Problem:

Write a function that groups anagrams together from a given array of strings.

Question 8: Remove Falsy Values from Array.

Problem:

Write a function to remove all falsy values from an array.

Falsy values include: false, 0, "", null, undefined, NaN

Question 9: Convert Array to Object.

Problem:

Write a function to convert an array of key-value pairs into an object.

Question 10: Find Missing Number.

Problem:

Given an array containing numbers from 1 to n with one number missing, write a function to find the missing number.

SOLUTION:

CODE: [Index.html]

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Web Assignment</title>
```

```
  <link
```

```
    href="https://fonts.googleapis.com/css2?family=Orbitron:wght@500&family=Poppins:wght@300;400&display=swap"
    rel="stylesheet">
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

```
  <h1>JS Problem Solver</h1>
```

```
  <select id="question">
```

```
    <option value="1">Reverse Words</option>
```

```
    <option value="2">Find Duplicates</option>
```

```
<option value="3">Character Frequency</option>

<option value="4">Merge Objects</option>

<option value="5">Flatten Array</option>

<option value="6">Longest Word</option>

<option value="7">Group Anagrams</option>

<option value="8">Remove Falsy</option>

<option value="9">Array to Object</option>

<option value="10">Find Missing Number</option>

</select>

<input type="text" id="input1" placeholder="Enter input">

<input type="text" id="input2" placeholder="Second input (if needed)">

<button onclick="solve()">Solve</button>

<div id="result"></div>

</div>

<script src="script.js"></script>

</body>

</html>
```

[Style.css]:

```
body { margin: 0; height:
100vh; display: flex;
justify-content: center;
align-items: center;
background-color: #b7ccd1;
```

```
font-family: 'Poppins', sans-  
serif;  
  
}
```

```
.card {      background-color: #ececec;  
  
padding: 45px 40px;   border-radius: 20px;  
  
width: 380px;   text-align: center;   box-  
shadow: 0 12px 30px rgba(0,0,0,0.18);  
  
}
```

```
h1 {  font-family: 'Orbitron', sans-  
serif;  color: #118c8f;  font-size:  
26px;  margin-bottom: 25px;  
  
}
```

```
select, input {      width:  
  
100%;      padding: 12px;  
  
border-radius:      12px;  
  
border: 2px solid #ff5a5a;  
  
margin-bottom:      15px;  
  
font-size: 14px;      outline:  
  
none;  
  
}
```

```
button {  background: linear-gradient(90deg, #ff7a6b,
#ff5a5a);  border: none;  padding: 13px 30px;
border-radius: 30px;  color: white;  font-size: 14px;
cursor: pointer;
}
```

```
#result {  margin-
top: 20px;  font-
size: 14px;
}
```

[Script.js]:

```
function solve() {  let q =
document.getElementById("question").value;  let
input1 = document.getElementById("input1").value;  let
input2 = document.getElementById("input2").value;  let
result;  switch (q) {

    // Q.1 Reverse Words

    case "1":

        result = input1.split(" ").map(w => w.split("").reverse().join("")).join(" ");

        break;

    // Q.2 Find Duplicates

    case "2":
```

```
    let arr = input1.split(",").map(Number);    result =  
arr.filter((item, index) => arr.indexOf(item) !== index);    result =  
[...new Set(result)];    break;
```

// Q.3 Character Frequency

```
case "3":  
  
    let freq = {};    for (let ch of input1) freq[ch]  
= (freq[ch] || 0) + 1;    result = freq;    break;
```

// Q.4 Merge Objects

```
case "4":  
  
    let obj1 = parseObj(input1);  
  
    let obj2 = parseObj(input2);  
result = { ...obj1 };    for (let k in obj2)  
result[k] = (result[k] || 0) + obj2[k];  
break;
```

// Q.5 Flatten Array

```
case "5":  
  
    let flatArr = eval("[ " + input1 + " ]");  
result = flatArr.flat(1);    break;
```

// Q.6 Longest Word

```
case "6":
```

```
result = input1.split(" ").reduce((a, b) => a.length > b.length ? a : b, "");
```

```
break;
```

```
// Q.7 Group Anagrams
```

```
case "7":
```

```
    let words = input1.split(",");
```

```
    let map = {};    words.forEach(w => {
```

```
    let key = w.split("").sort().join("");
```

```
    if (!map[key]) map[key] = [];
```

```
    map[key].push(w);
```

```
    });
```

```
    result = Object.values(map);
```

```
    break;
```

```
// Q.8 Remove Falsy
```

```
case "8":
```

```
    let arr2 = input1.split(",");
```

```
    result = arr2.filter(v => v && v !== "0" && v !== "false" && v !== "null" && v !== "undefined" && v !== "NaN");
```

```
    break;
```

```
// Q.9 Array to Object
```

```
case "9":
```

```

        let pairs = input1.split(",");
result = {};      pairs.forEach(p
=> {          let [k, v] =
p.split(":");      result[k] = v;

        });
break;

```

```

// Q.10 Missing Number

case "10":

    let nums = input1.split(",").map(Number);

    let n = nums.length + 1;      let sum = n *
(n + 1) / 2;      let actual = nums.reduce((a, b)
=> a + b, 0);      result = sum - actual;

break;

}

document.getElementById("result").innerText = JSON.stringify(result);
}

function parseObj(str) {

    let    obj    =    {};

    str.split(",").forEach(p => {

    let  [k,  v]  =  p.split(":");

    obj[k] = Number(v);

    });

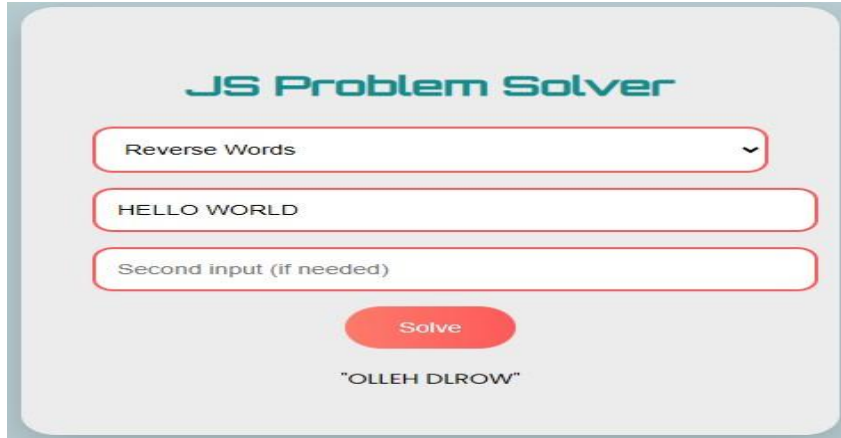
    return obj;

```


}

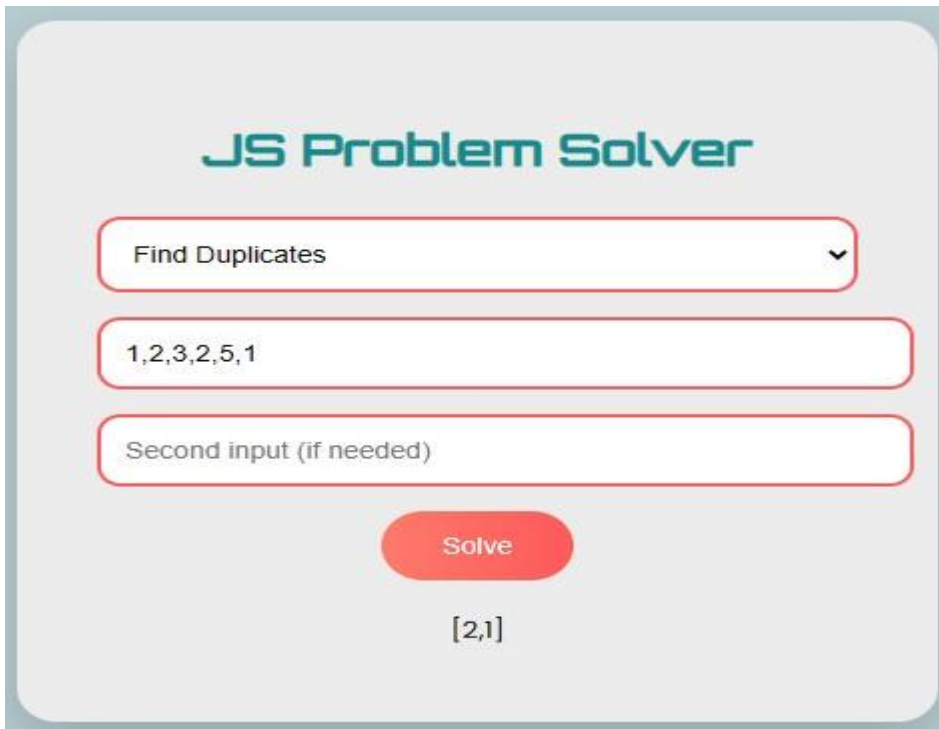
[OUTPUT]:

Sol 1: Reverse Word.



The image shows a web-based interface for a JavaScript problem solver. At the top, it says "JS Problem Solver" in a teal font. Below this is a dropdown menu with "Reverse Words" selected. Underneath the dropdown is a text input field containing "HELLO WORLD". Below that is another text input field labeled "Second input (if needed)". A red "Solve" button is positioned below the input fields. At the bottom, the output is displayed as "OLLEH DLROW" in quotes.

Sol 2: Find Duplicates.



The image shows a web-based interface for a JavaScript problem solver. At the top, it says "JS Problem Solver" in a teal font. Below this is a dropdown menu with "Find Duplicates" selected. Underneath the dropdown is a text input field containing "1,2,3,2,5,1". Below that is another text input field labeled "Second input (if needed)". A red "Solve" button is positioned below the input fields. At the bottom, the output is displayed as "[2,1]" in a monospace font.

Sol 3: Character Frequency.

JS Problem Solver

Character Frequency

DIVESH KINHA

Second input (if needed)

Solve

```
{ "D":1,"I":2,"V":1,"E":1,"S":1,"H":2," ":1,"K":1,"N":1,"A":1 }
```

Sol 4: Merge Objects.

JS Problem Solver

Merge Objects

a:1,b:5

b:7,c:9

Solve

```
{ "a":1,"b":12,"c":9 }
```

Sol 5: Flatten Array.

JS Problem Solver

Flatten Array ▼

1,[2,3],[4,5]

Second input (if needed)

Solve

[1,2,3,4,5]

Sol 6: Longest Word.

JS Problem Solver

Longest Word ▼

I believe in god's plan

Second input (if needed)

Solve

"believe"

Sol 7: Group Anagrams.

JS Problem Solver

Group Anagrams

Eat,Sleep,Repeat

Second input (if needed)

Solve

[["Eat"],["Sleep"],["Repeat"]]

Sol 8: Remove falsy.

JS Problem Solver

Remove Falsy

0,1,true,3

Second input (if needed)

Solve

["1","true","3"]

Sol 9: Array to Object.

Sol 10: Find Missing No.

JS Problem Solver

Array to Object



["a",1]

["b",2]

Solve

{}

JS Problem Solver

Find Missing Number



1,2,4,5,6

Second input (if needed)

Solve

3